Complete pH & Acid-Base Equations (Lectures 1–5)

1. Basic Definitions

pH = -log[H+]

pOH = -log[OH-]

pH + pOH = 14

Example: If [H+] = 10^-3 → pH = 3

2. Ion Product of Water

Kw = [H+][OH-] = 10^-14

Example: If [H+] = 10^-5 → [OH-] = 10^-9

3. Strong Acids

Fully dissociated: [H+] = concentration

pH = -log[H+]

Example: 0.1 M HCl → pH = 1

4. Strong Bases

pOH = -log[OH-]

pH = 14 - pOH

Example: 0.01 M NaOH → pH = 12

5. Weak Acids

pH = 1/2 pKa + 1/2 pCa

Example: pKa = 4.76, Ca = 0.1 → pH = 2.88

6. Weak Bases

pH = 14 - 1/2 pKb - 1/2 pCb

Example: pKb = 4.76, Cb = 0.1 → pH = 11.12

7. Buffers (Henderson-Hasselbalch)

Acidic buffer: pH = pKa + log(salt/acid) => [salt]/ [acid] [ ] => refers to concentration

Example: equal concentrations → pH = pKa

Basic buffer: pH = 14 - pKb - log(salt/base)

Example: NH_4OH/NH_4Cl → pH = 8.24

8. Salt Hydrolysis

Strong acid + strong base → pH = 7

Strong acid + weak base → acidic

Weak acid + strong base → basic

Weak acid + weak base → depends on Ka & Kb

9. Indicators

pH = pKIn ± 1 pkIn => means pk Indicator

Example: Phenolphthalein range = 8.3 – 10

10. Buffer Capacity

Maximum when [salt] = [acid]

pH = pKa ± 1

11. Titration Key Equations

Strong acid–strong base → pH = 7 at equivalence

Weak acid–strong base → pH > 7

Weak base–strong acid → pH < 7